IMS LMS Functionality Update

Implemented modules, security controls, automated validation and remaining external setup

Area	Current status
Learning lifecycle	Catalog, free enrollment, paid checkout workflow, player progress and course certificates are implemented.
Student operations	Tasks, projects, assessments, reviews, jobs, attendance, calendar, to-dos, tickets and notifications are available.
Admin operations	Learning management, course sales, certificate issue flow, public inquiries and learning Excel export are available.
Security and testing	Ownership and role-based access checks are implemented and regression-tested in an isolated database.

1. Public Learning and Enrollment

Public course catalog

Published courses are exposed without private lesson resources.

- Search by course, skill and domain.
- Domain, free/paid and trending filters.
- Sort by newest, price and duration.
- Price and discount display on catalog cards.

Course detail and access

The course detail page protects non-preview lesson content.

- Preview lessons remain visible to visitors.
- Free courses enroll a logged-in student immediately.
- Paid courses create a purchase record before access is granted.
- Student continues learning from My Learning.

Razorpay course workflow

A signed Razorpay verification path activates the enrollment only after successful payment.

- Course purchases are tracked separately from project fees.
- Course Sales screen shows paid and pending records.
- Webhook handler receives the raw request body for signature validation.

2. Learning, Certificates and Student Workflows

Learning player and progress

Admin-created modules and lessons are shown to enrolled students.

- Lesson completion updates progress percentage.
- Completion is recorded at 100%.
- Admin can assign or revoke enrollment.
- Enrollment list shows student progress and source.

Certificate milestones

Project and learning certificate eligibility are calculated server-side.

- Project milestones: 75% progress, 95% completion, 100% LOR.
- Learning milestones: 75% progress and 100% completion.
- Admin generates eligible certificates only.
- Student page shows locked, eligible and issued states.

Assessments, project review and job flow

Assessment history, project review and job application operations are implemented.

- Assessment answers are graded server-side; correct answers are hidden before submit.
- A student can review only an assigned project.
- Job application unlocks only when assigned project progress reaches 100%.

3. Admin, Support and Communication

Admin learning operations

Admin controls course status, lessons, enrollment, price and discount.

- Learning Courses and Course Editor pages.
- Question bank, project review and job application screens.
- Course Sales screen for purchase records.
- Public inquiry inbox with status management.

Support tickets and notifications

Support operations are scoped to the correct student.

- Students see and reply only to their own tickets.
- Admins can reply and resolve tickets.
- Admin reply creates an in-app student notification.
- Student reply creates in-app notifications for admins.
- A user cannot mark another user notification as read.

Operational email

Brevo API delivery support was added using configured mail settings.

- A real delivery test was sent to the approved test inbox.
- Email failures are handled without exposing secrets.
- Certificate email will work after PDF upload storage is configured.

4. Security and Automated Regression Testing

Strict ownership checks

Access checks were hardened in project, team, task, ticket and notification routes.

- Student cannot open another student project or team.
- Student task list is limited to assigned tasks.
- Team calendar access requires team membership.
- Personal to-dos are bound to the signed-in user.

Regression suite

A script creates its own test admin, students, teams, projects, tasks and course in an isolated MongoDB database.

- Free enrollment and My Learning verification.
- Project, team, task, ticket and notification denial checks.
- Attendance check-in/out, calendar and to-do checks.
- Assessment grading, review ownership, job unlock and admin-only route checks.
- Test database is deleted after completion; live LMS data is not used.

5. Reporting and External Setup Remaining

Learning Progress and Certificates Excel report

Admin Reports now has a real download for learning operations.

- Student, course, domain and progress.
- Enrollment source and paid amount.
- 75% and completion certificate IDs.
- Last updated timestamp.

External configuration still required

These items need credentials or dashboard action and were intentionally not faked.

- Cloudinary cloud name, API key and secret for live PDF certificate upload/download test.
- Razorpay Dashboard webhook URL and a real payment test, currently deferred.
- Legacy live LMS data export or database access for migration, currently deferred.
- Landing-page visual work such as FAQ/testimonials/newsletter is being handled separately.

Validation note: The core workflow regression suite passed after the latest changes. Paid Razorpay checkout and Cloudinary-backed certificate generation require their external production setup before live testing.